

L_RAG Presentation Slide Modifications Report

Text Mining Final Project Presentation Updates

September 18, 2026

Summary of Modifications

This document summarizes the insertion of two new detailed architecture slides into the presentation deck `Text_Mining.pdf` following Slide 5. All changes align with the official ablation report (`ablation_report.pdf`) and incorporate feedback regarding Knowledge Graph schema construction and multi-stage pipeline breakdown.

Slide #	Action	Summary of Change
Slide 6	INSERTED (NEW)	L_RAG Pipeline: Stage-by-Stage Processing: Breaks down the 4 stages (Indexing, KG Overlay Joiner, RRF Fusion & Rerank, Grounded LLM Gen).
Slide 7	INSERTED (NEW)	G-LRAG Knowledge Graph: Schema & Construction: Details node hierarchy (Docs, Provisions, Chunks), edge types, and ingested scale (151k docs, 1.38M provisions, 1.51M chunks).

Table 1: Summary table of inserted presentation slides.

Detailed Breakdown of Inserted Slides

1. Slide 6: Stage-by-Stage Pipeline Processing

- **Stage 1 (Indexing):** Metadata enrichment (`Title || Header || Passage`) + underthesesa BM25 term tokenization.
- **Stage 2 (KG Overlay):** Dynamic `id_str_filter` validity masking + 1-hop `CHUNK_NEXT` reading-order expansion.
- **Stage 3 (Fusion & Rerank):** RRF ($k = 60$) candidate fusion ($K = 30$) + GPU load-run-unload Cross-Encoder reranking ($N = 10$).
- **Stage 4 (Generation):** `gpt-4o-mini` ($T = 0.0$) with statutory citation grounding and `INSUFFICIENT_CONTEXT` abstention.

2. Slide 7: Knowledge Graph Schema and Construction

- **Graph Node Hierarchy:** Formalizes Document (D), Provision (P), and Chunk (C) entity nodes.
- **Edge Schema:** Distinguishes Structural edges (`DOC_HAS_PROVISION`, `PROVISION_HAS_CHUNK`, `CHUNK_NEXT`, `PROVISION_NEXT`) from Legal Relations (`amend`, `replace`, `cite`, `repeal`).
- **Ingested Scale Table:** Embeds formal corpus statistics (151,624 Documents, 1,386,267 Provisions, 1,513,376 Chunks, 883,256 Document Edges).
- **Validity Rule:** Highlights dynamic `as_of_date` whitelisting and quarantining of unverified records.